

2016.0 RANGE ROVER (LG), 303-01

ENGINE - TDV6 3.0L DIESEL - GEN 2/TDV6 3.0L DIESEL

OIL PUMP (G1509152)

REMOVAL AND INSTALLATION

12.60.26	PUMP - OIL - RENEW	3000 CC, TDV6	4.5	USED WITHINS	+
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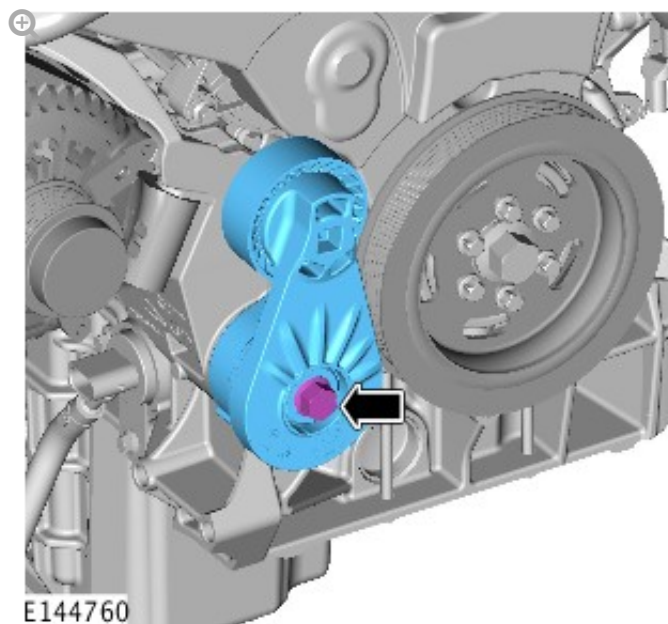
PART(S)

STEP	REPLACE PART / RENEW PART	PART NAME
Installation Step 15	Renew Part	Crankshaft pulley bolt

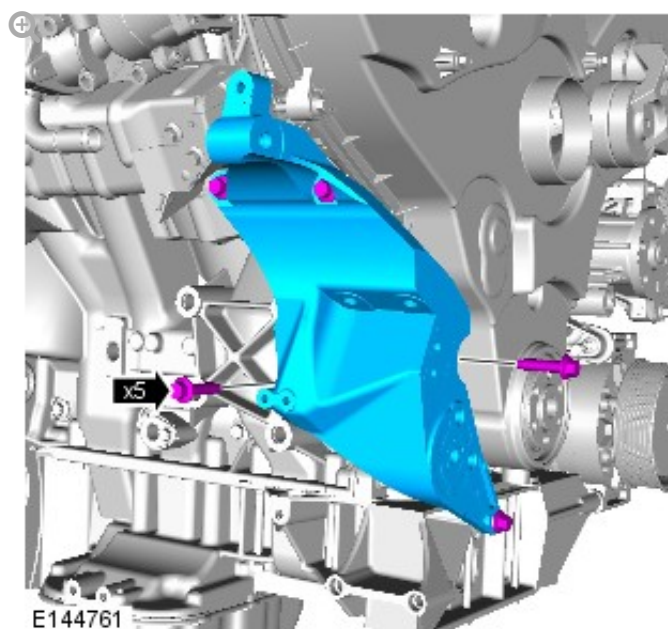
REMOVAL

1. Disconnect the battery ground cable.
Refer to: [Specifications](#) (414-00 Battery and Charging System - General Information, Specifications).
2. Remove the generator.
Refer to: [Generator](#) (414-02A Generator and Regulator - TDV6 3.0L Diesel, Removal and Installation).

3.

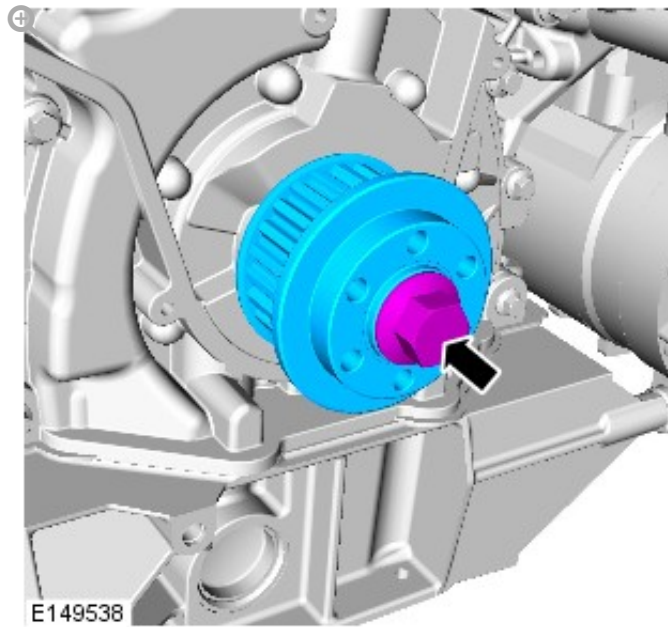


4.



5. Refer to: [Timing Belt](#) (303-01F Engine - SDV6 3.0L Diesel - Hybrid Electric Vehicle, Removal and Installation).

6.



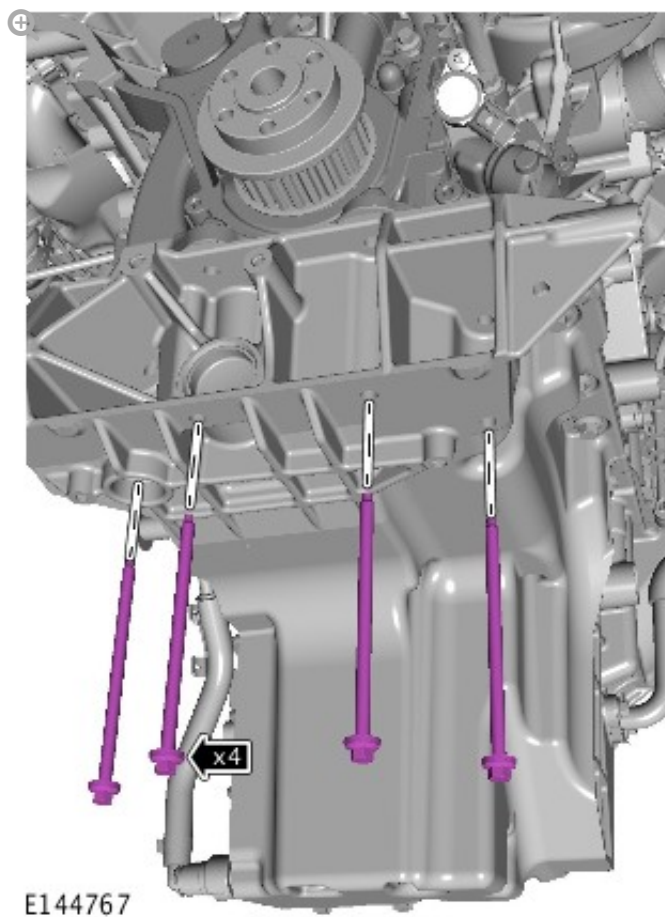
7.

WARNING:

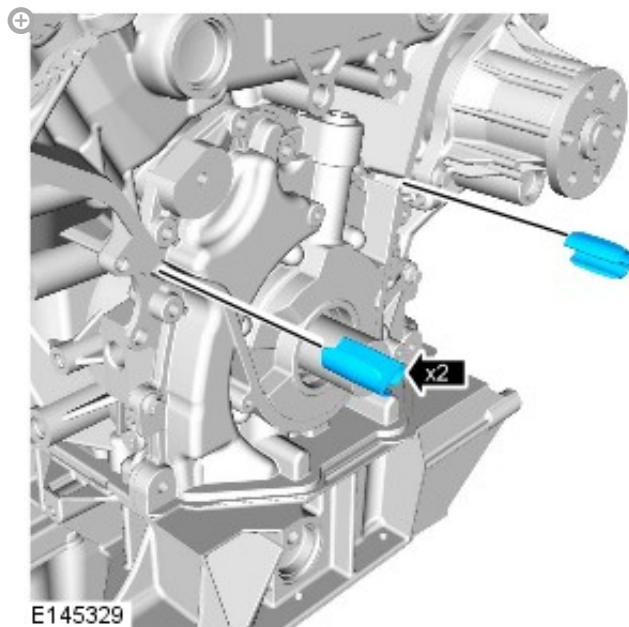
Do not work on or under a vehicle supported only by a jack.
Always support the vehicle on safety stands.

Raise and support the vehicle.

8.

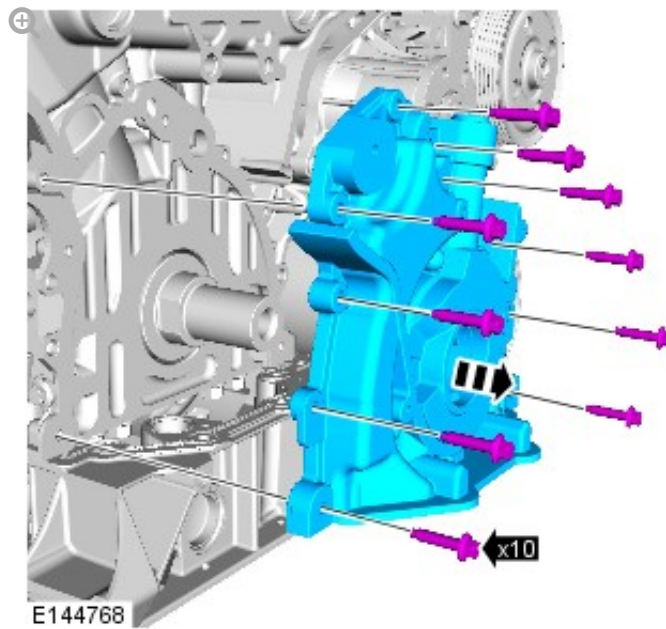


9.

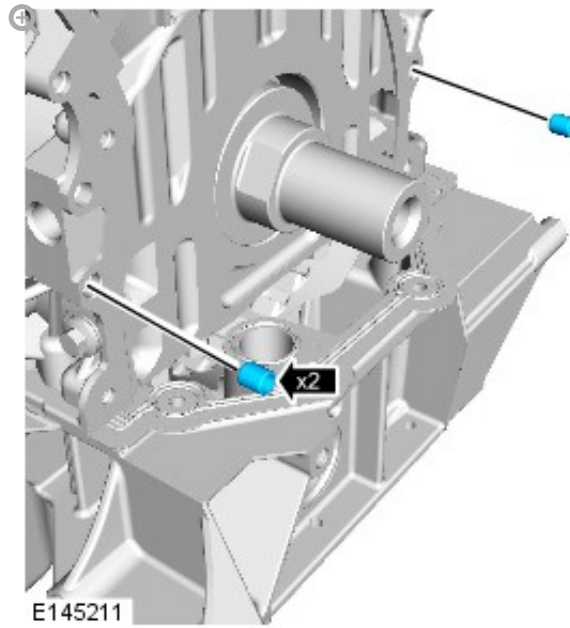


Remove the timing belt cover sealing strips.

10.



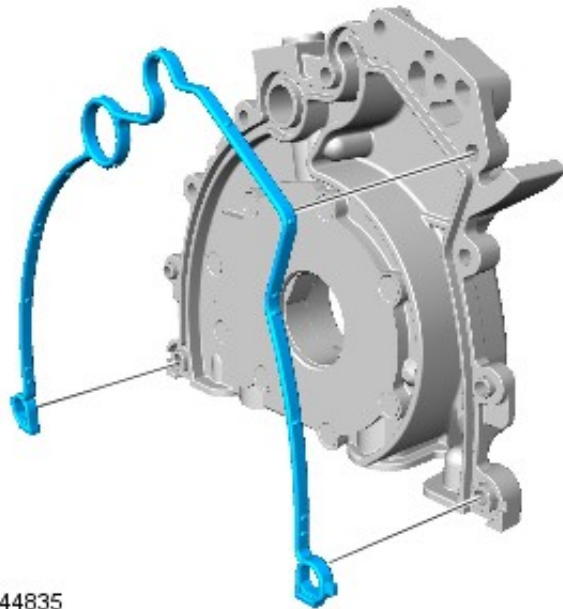
11.



Remove the alignment dowels (if equipped).

INSTALLATION

1.

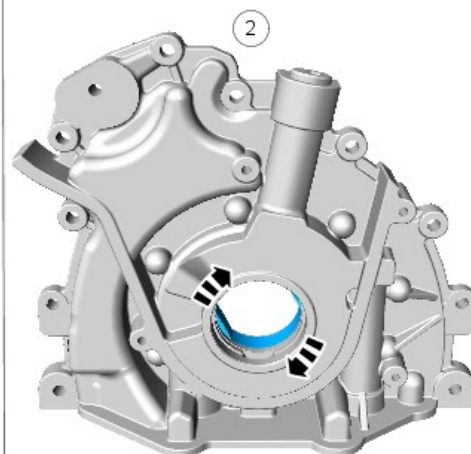
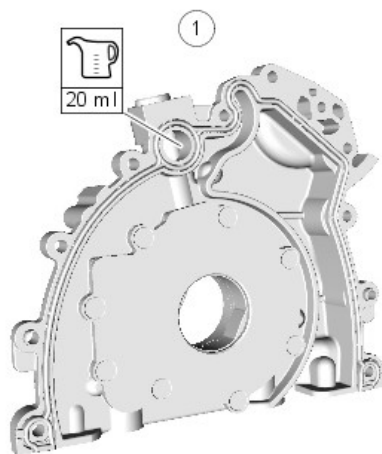


E144835

2.

NOTE:

Prime the oil pump.



E144763

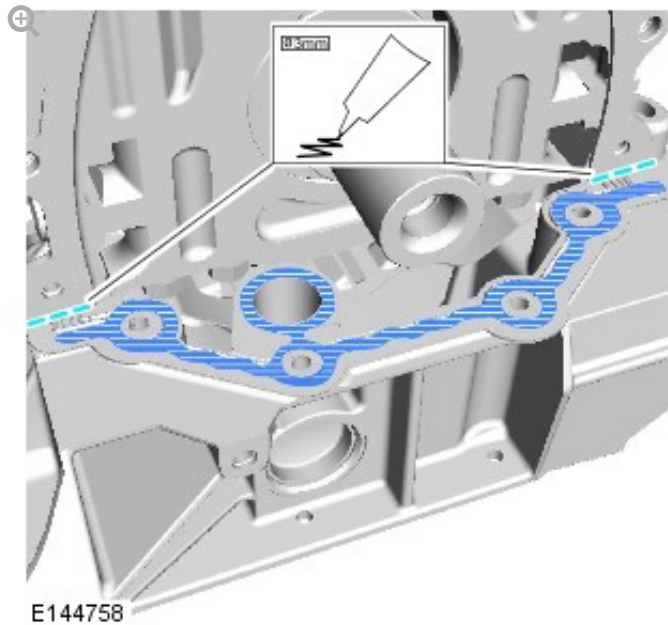
1. Fill the orifice shown with 20 ml of engine oil.

2. Rotate the oil pump drive 2 complete turns.

3.

CAUTION:

Make sure that the mating faces are clean and free of foreign material.



Apply sealant WSE-M4G323-A5 in an 3mm bead to the two areas shown, also apply a smear of sealant to the gasket face as illustrated.

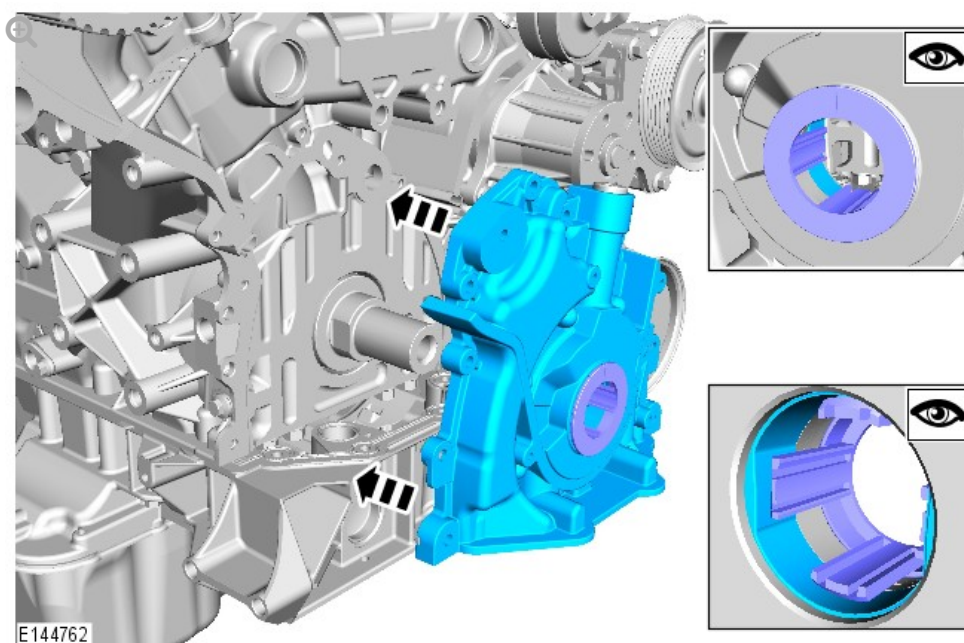
4.

CAUTION:

Make sure that the mating faces are clean and free of foreign material.

NOTE:

Make sure the gasket is installed correctly.

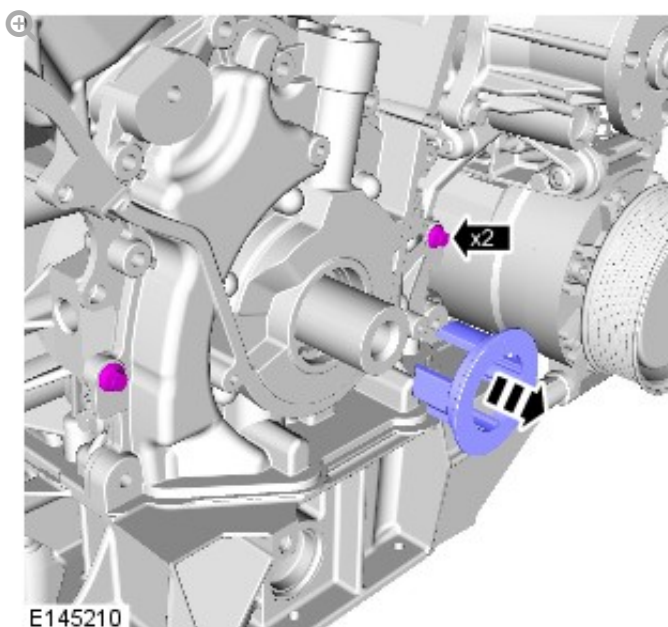


Install the oil pump.

5.

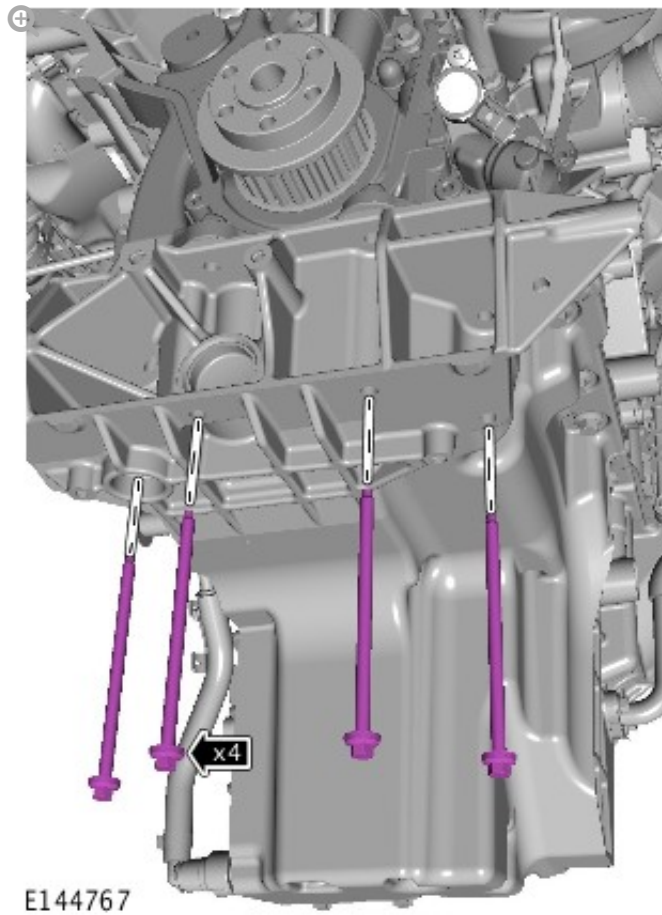
NOTE:

Remove the oil pump locator.



Install two bolts to the oil pump, only finger tighten at this stage.

6.



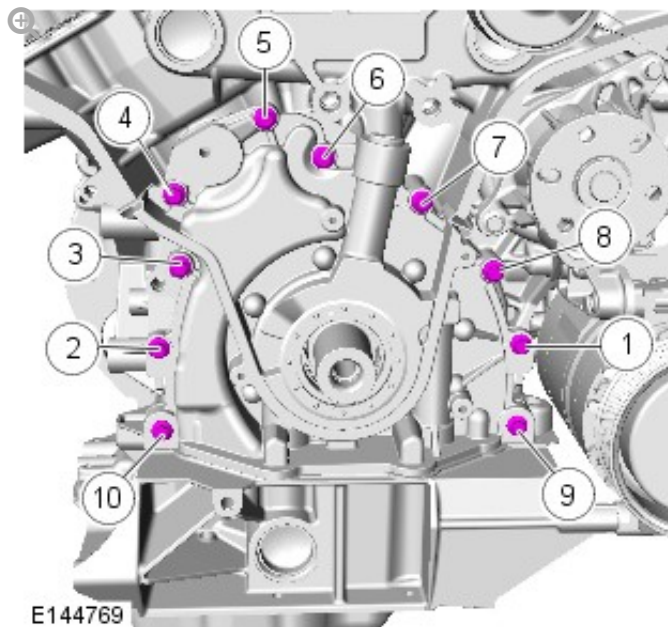
Install the oil pump lower retaining bolts.

Torque: 10 Nm

7.

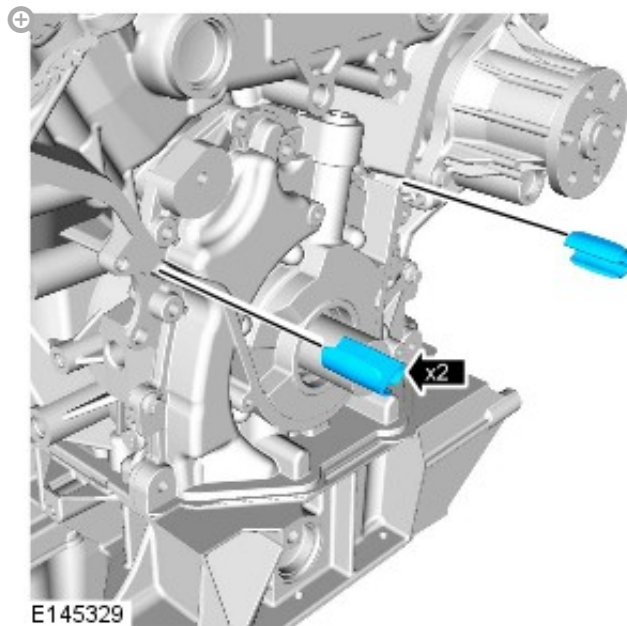
NOTE:

Tighten the bolts in the sequence shown to 10 Nm .



Secure the oil pump.

8.



Install the timing belt cover sealing strips.

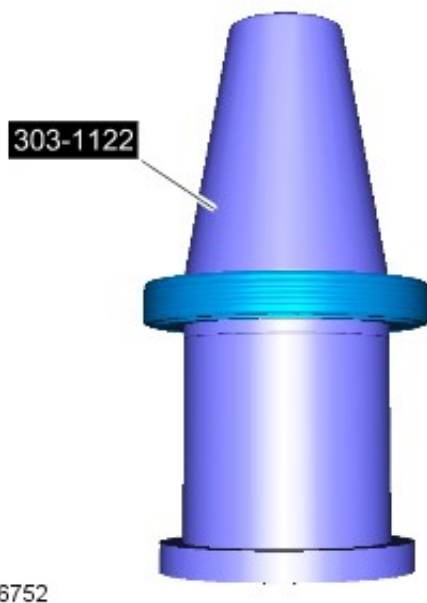
9.

CAUTION:

Do not use any lubricant on the crankshaft front seal, special tools or the crankshaft. Failure to follow this instruction may result in damage to the vehicle.

NOTE:

Make sure all component mating faces are clean.



Install a new crankshaft front seal to the special tool.

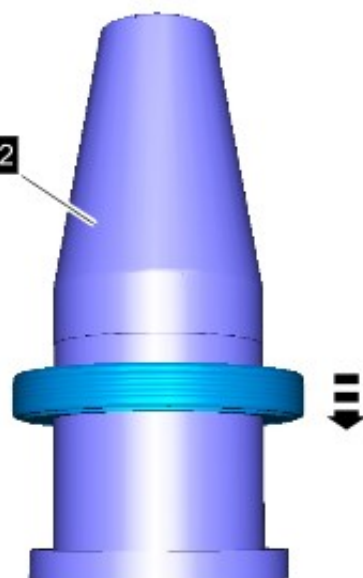
10.

NOTE:

Remove the sleeve from the special tool.



303-1122



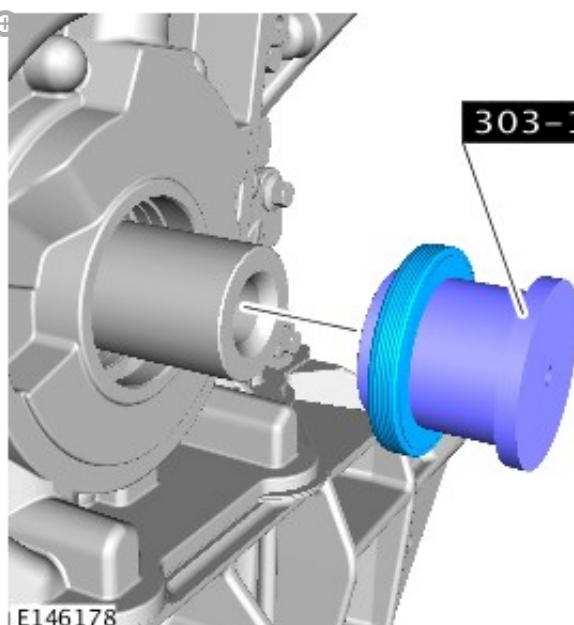
E116753

Reposition the crankshaft front seal along the special tool.

11.



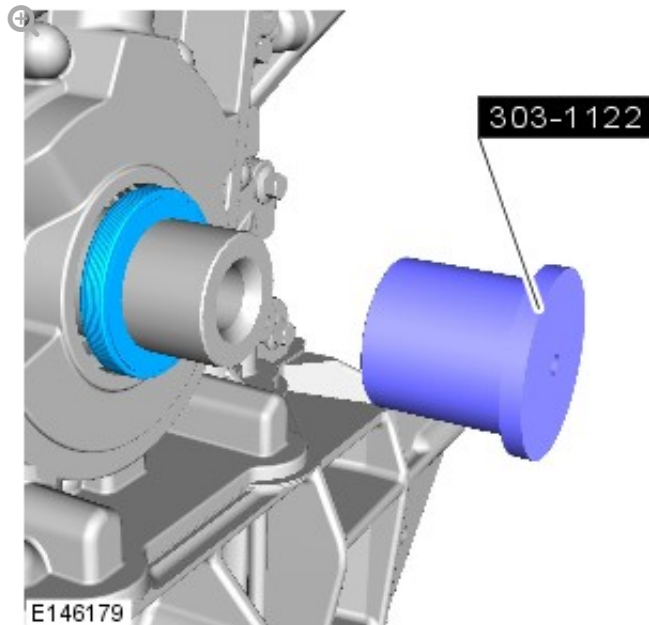
303-112



E146178

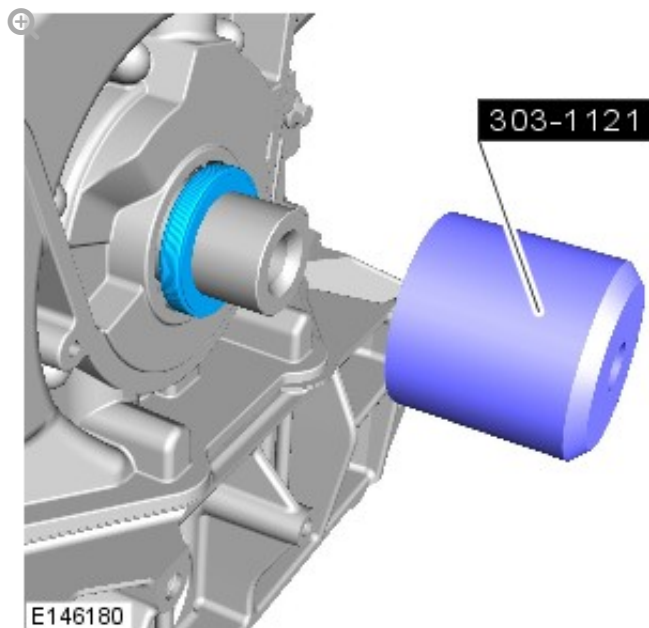
Install the special tool to the crankshaft.

12.



Remove the special tool from the crankshaft.

13.

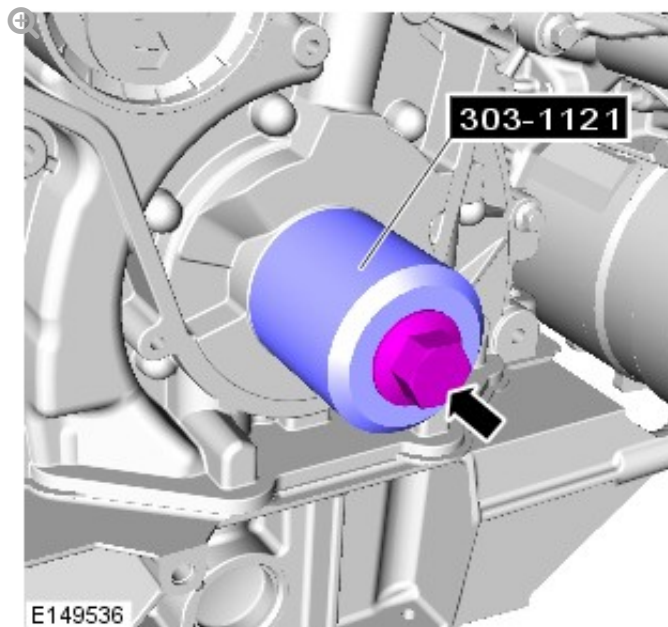


Install the special tool to the crankshaft.

14.

NOTES:

- Make sure the seal is installed correctly.
- Use the crankshaft bolt with the service tool.

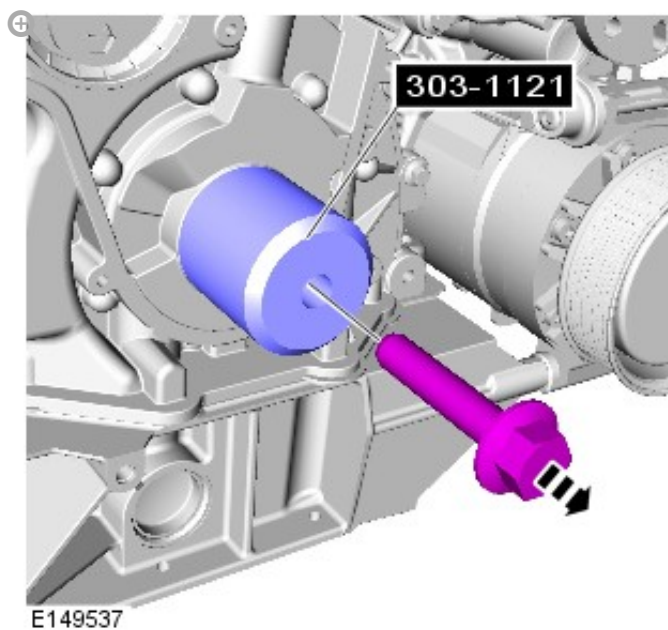


Install the special tool to the crankshaft.

15.

NOTES:

- Make sure the seal is installed correctly.
- Wait 10 seconds before removing the crankshaft seal installation tool.



Remove and discard the crankshaft pulley retaining bolt.

Renew Part: [Crankshaft pulley bolt](#).

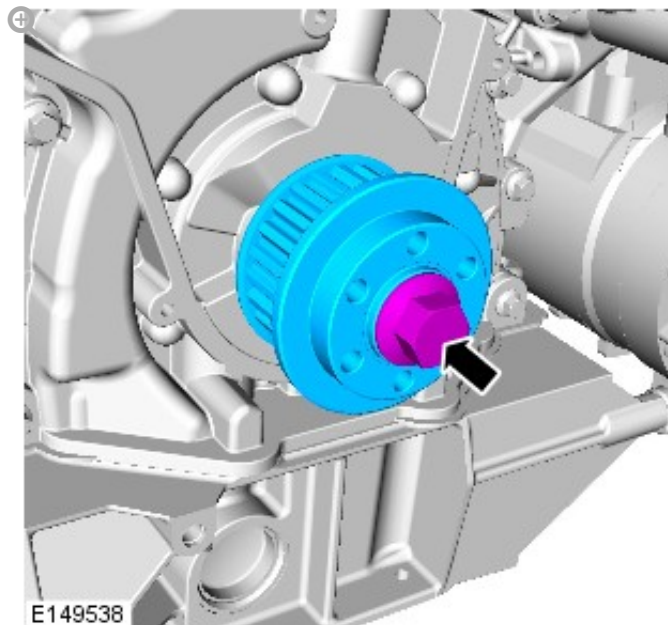
16.

CAUTION:

Make sure that a new crankshaft pulley bolt is installed.

NOTE:

Before installing the crankshaft pulley retaining bolt, note the thread size of the bolt (there are two different sized threads, M14 and M16). Failure to follow this instruction may result in damage to the vehicle.



Install the crankshaft pulley.

Torque:

Tighten the M14 retaining bolt in two stages:

Stage 1: **140 Nm**

Stage 2: **90°**

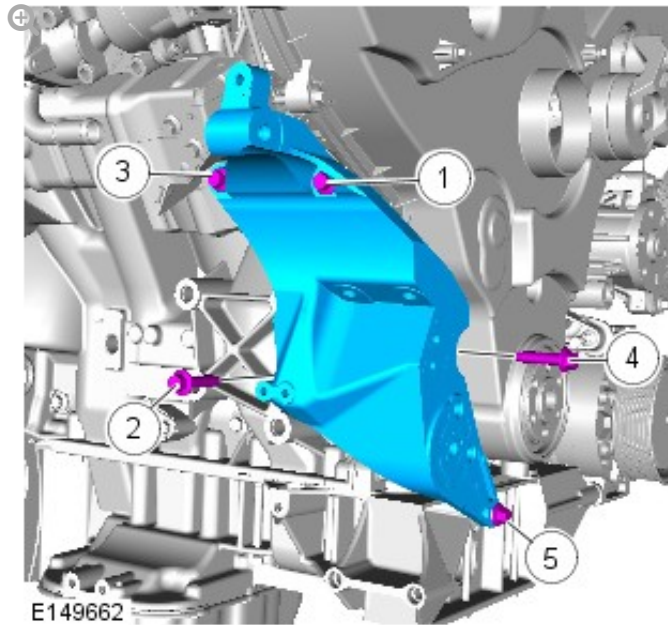
Tighten the M16 retaining bolt in three stages:

Stage 1: 100 Nm

Stage 2: 35°

17. Refer to: [Timing Belt](#) (303-01F Engine - SDV6 3.0L Diesel - Hybrid Electric Vehicle, Removal and Installation).

18.

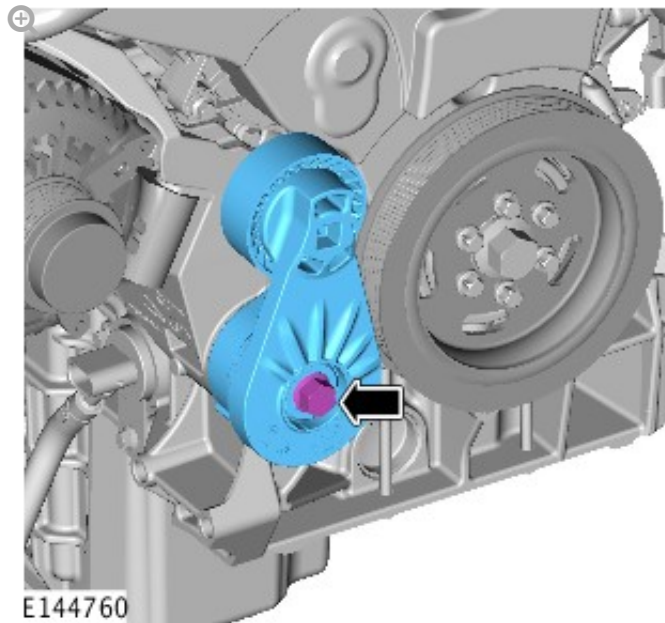


Tighten the bolts in the sequence shown.

Torque: 22 Nm

19. Remove the generator.
Refer to: [Generator](#) (414-02A Generator and Regulator - TDV6 3.0L Diesel, Removal and Installation).

20.



Torque: 45 Nm

21. Connect the battery ground cable.
Refer to: [Specifications](#) (414-00 Battery and Charging System - General Information, Specifications).